

LAB 02

Internal Microservice Catalog and Health Portal

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1. Objective

To convert the functional requirements from Lab 01 into Agile Epics and User Stories, prioritize and estimate the backlog using Story Points, simulate Scrum sprints in Jira, and analyse sprint progress using the Burndown Chart.

2. Agile Backlog

The Lab 01 requirements were converted into five Epics and twelve User Stories. The User Stories were prioritized and estimated using Fibonacci-based Story Points.

Epic	User Stories	Total Story Points
IMCHP-1 – Microservice Health Monitoring	IMCHP-6, IMCHP-7, IMCHP-8	18
IMCHP-2 – Microservice Catalog	IMCHP-9, IMCHP-10	6
IMCHP-3 – Dependency Management	IMCHP-11, IMCHP-12	13
IMCHP-4 – API Documentation	IMCHP-13, IMCHP-14	6
IMCHP-5 – Downtime & Alert Management	IMCHP-15, IMCHP-16, IMCHP-17	13

Total backlog estimate: 56 Story Points.

3. User Story and Story Point Summary

Key	User Story	Priority	Story Points
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IMCHP-6	Monitor Microservice Health	High	8
IMCHP-7	Track Health Check Results	High	5
IMCHP-8	Calculate Service Availability	High	5
IMCHP-9	View Microservice Catalog	High	3
IMCHP-10	View Service Information	High	3
IMCHP-11	View Dependency Graph	High	8
IMCHP-12	Explore Service Dependencies	High	5
IMCHP-13	View API Documentation	Medium	3
IMCHP-14	View Service API Details	Medium	3
IMCHP-15	Record Downtime Event	High	5
IMCHP-16	Receive Downtime Alert	High	5
IMCHP-17	View Downtime History	High	3

Story Points use the Fibonacci-style scale recommended in the lab handout (for example, 3, 5 and 8) to represent relative effort, complexity and uncertainty.

4. Sprint Execution

Sprint	Planned Stories	Story Points	Result
Sprint 1	5 User Stories	24	Completed
Sprint 2	7 User Stories	32	Completed

The sprint workflow was simulated using To Do → In Progress → Done. Sprint 1 was completed with 24 Story Points and Sprint 2 was completed with 32 Story Points. Together, the two sprints completed all 56 Story Points in the backlog.

5. Reflection / Write-up

1. Did your estimations reflect the actual effort?

The estimates were reasonable for relative effort because larger and more complex stories were assigned higher Story Points, while simpler viewing and documentation stories received lower estimates. Because the activity was a simulated sprint, actual software-development effort was not measured directly. The estimates were nevertheless appropriate for planning and comparing the difficulty of the stories.

2. Was your backlog well-prioritized?

Yes. The backlog was prioritized according to business importance and user needs. Core monitoring, service availability, catalog, dependency, and downtime functionality received high priority, while API documentation was given medium priority. This helped determine which work should be handled first.

3. How did your simulated sprint align with your plan?

The simulation followed the planned Scrum workflow. Stories were selected from the backlog, estimated with Story Points, and moved through To Do, In Progress and Done. Sprint 1 completed 5 stories worth 24 points and Sprint 2 completed the remaining 7 stories worth 32 points. Therefore, all planned User Stories were completed across the two sprints.

4. What insights did the burndown chart give about your team's capacity?

The Burndown Chart showed the remaining Story Points decreasing as work was completed. Sprint 2 began with 32 Story Points and reached zero remaining work by the end of the simulation. This indicates that the planned Sprint 2

workload was completed and provides a useful baseline for estimating future sprint capacity.

5. Screenshots

Work	Assignee	Reporter	Story Points	Priority	Status	Resolution	Created
IMCHP-1: Microservice Health Monitoring	Unassigned	Abhishek Dey	None	High	To Do	Unresolved	30 Aug 2025, 21:14
IMCHP-2: Track Health Check Results	Unassigned	Abhishek Dey	3	High	Done	Unresolved	30 Aug 2025, 21:25
IMCHP-3: Calculate Service Availability	Unassigned	Abhishek Dey	3	High	Done	Unresolved	30 Aug 2025, 21:25
IMCHP-4: Microservice Catalog	Unassigned	Abhishek Dey	None	High	To Do	Unresolved	30 Aug 2025, 21:18
IMCHP-5: View Microservice Catalog	Unassigned	Abhishek Dey	3	High	Done	Unresolved	30 Aug 2025, 21:28
IMCHP-6: View Service Information	Unassigned	Abhishek Dey	3	High	Done	Unresolved	30 Aug 2025, 21:27
IMCHP-7: Dependency Management	Unassigned	Abhishek Dey	None	High	To Do	Unresolved	30 Aug 2025, 21:19
IMCHP-8: Explore Service Dependencies	Unassigned	Abhishek Dey	3	High	Done	Unresolved	30 Aug 2025, 21:28
IMCHP-9: View Dependency Graph	Unassigned	Abhishek Dey	3	High	Done	Unresolved	30 Aug 2025, 21:27
IMCHP-10: API Documentation	Unassigned	Abhishek Dey	None	Medium	To Do	Unresolved	30 Aug 2025, 21:20
IMCHP-11: View API Documentation	Unassigned	Abhishek Dey	3	Medium	Done	Unresolved	30 Aug 2025, 21:28

Figure 1 – Jira Backlog with Epics and User Stories

Story Point	Priority	Status	Description	Created	Updated	Done
IMCHP-1	High	To Do	Microservice Health Monitoring	30 Aug 2026, 21:14	30 Aug 2026, 22:38	None
IMCHP-2	High	To Do	Track Health Check Results	30 Aug 2026, 21:25	30 Aug 2026, 22:37	None
IMCHP-3	High	To Do	Calculate Service Availability	30 Aug 2026, 21:29	30 Aug 2026, 22:36	None
IMCHP-4	High	To Do	Microservice Catalog	30 Aug 2026, 21:38	30 Aug 2026, 22:38	None
IMCHP-5	High	To Do	View Microservice Catalog	30 Aug 2026, 21:46	30 Aug 2026, 22:38	None
IMCHP-6	High	To Do	View Service Information	30 Aug 2026, 21:57	30 Aug 2026, 22:37	None
IMCHP-7	High	To Do	Dependency Management	30 Aug 2026, 21:59	30 Aug 2026, 22:38	None
IMCHP-8	High	To Do	Explore Service Dependencies	30 Aug 2026, 21:59	30 Aug 2026, 22:38	None
IMCHP-9	High	To Do	View Dependency Graph	30 Aug 2026, 21:57	30 Aug 2026, 22:38	None
IMCHP-10	Medium	To Do	API Documentation	30 Aug 2026, 21:20	30 Aug 2026, 22:35	None
IMCHP-11	Medium	To Do	View API Documentation	30 Aug 2026, 21:28	30 Aug 2026, 22:31	None
IMCHP-12	Medium	To Do	View Service API Details	30 Aug 2026, 21:29	30 Aug 2026, 22:23	None

Figure 2 – Story Point Assignments

Sprint Name	Status	Done
Monitor Microservice Health	In Progress	IMCHP-1
Track Health Check Results	In Progress	IMCHP-2
View Microservice Catalog	In Progress	IMCHP-3
View API Documentation	In Progress	IMCHP-4

Figure 3 – Sprint 1 Active Sprint Board

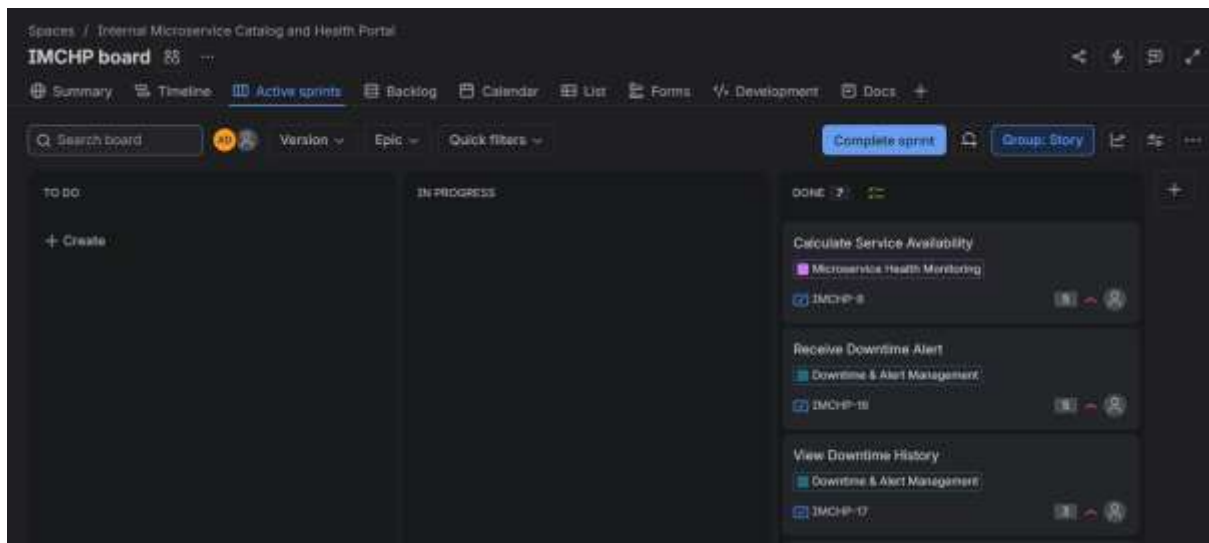


figure 4 – Sprint 2 Active Sprint Board

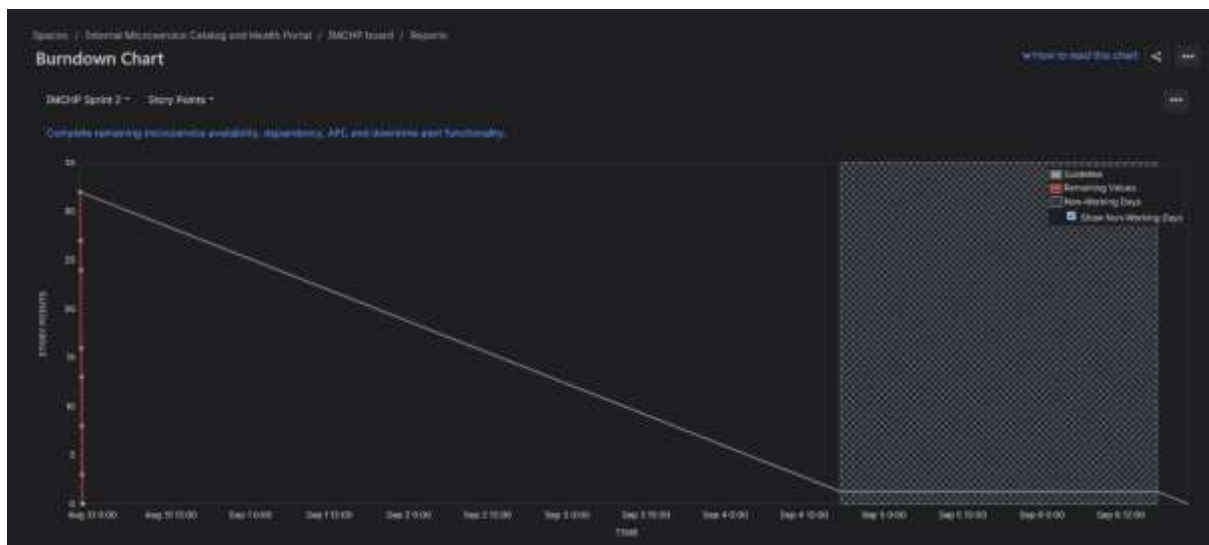


Figure 5 – Sprint 2 Burndown Chart

6. Conclusion

This lab demonstrated the use of Jira for Agile backlog management and Scrum sprint simulation. The Lab 01 requirements were transformed into Epics and User Stories, estimated using Story Points, planned across two sprints, and tracked using the Active Sprint board and Burndown Chart.

